# Math 260 Project: Modeling Gambling Addiction

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**Abstract**

According to the American Psychological Association, depending on country of study, approximately 0.4% to 2% of the total population among adults display symptoms of gambling addiction [1]. Despite being a highly profitable industry, gambling also poses problems to both individuals and the society. Therefore, we find it helpful to investigate the mechanisms of gambling addiction. This report proposes (1) an opponent process model based on [2] and (2) a Hebbian learning model [3] to simulate the neuroadaptive development of gambling addiction. By modeling the process of physiological changes and habituation, we derived qualitative results on the influences of different factors in inducing gambling addiction, and concluded that among all the changing parameters, fluctuation in monetary reward is the most significant stimulus in accelerating the addiction process.

## 1 Introduction

Gambling is a very successful commercial enterprise, yet many gamblers become addicted and face negative outcomes. According to the American Gaming Association, in the third quarter of 2023, total commercial gambling revenue has increased for about 6.7% compared to 2022, and reached a total of $16.26 billion [4]. On the other hand, the National Council on Problem Gambling estimated that around two million adults in the US are struggling with pathological gambling, while another four to six million are considered to have mild or moderate gambling problems [5].

Therefore, it is necessary and important to investigate the process of gambling addiction in order to provide more guidance in dealing with this pervasive issue. However, while gambling addiction is very widespread, our understanding about the addiction mechanism is still limited. In particular, most of the current studies of gambling addiction focus on proposing a theory about the biological reaction of the body's dopamine regulations towards the stimulus of gambling (see e.g. [6], [7]). Furthermore, there has not been much research conducted on modeling gambling addiction from a mathematical approach. While the previous models provided a comprehensive explanation of the observed characteristics of pathological gambling, the research literature has a shortage of simple mathematical models which provide good qualitative descriptions of the addiction process.

For our project, we propose and analyze models for the neurophysiology of gambling addiction inspired by [2] and [3]. These models provide a good qualitative description of the process of gambling addiction undergone by an individual gambler, and have complementary strengths. Furthermore, by introducing measures on the quantity of addiction level, we were able to qualitatively evaluate the effect of different factors. Specifically, we found that monetary reward is the major factor that induces the addiction process.

# 2 Background

## 2.1 To What Extent is Gambling an Addiction?

The conventional definition of addiction refers only to drug addiction [8]. Addictive substances intervene in the mesocorticolimbic pathways which govern dopamine release in response to primary rewards such as food, entertainment, and sex [2]. The effect of addictive drugs on the brain is similar to that of primary rewards, but larger in magnitude and much faster. While each drug type has a different operational mechanism, the common feature of many addictive drugs is a large increase in dopamine release, up to two to ten times more than food or sex. According to incentive sensitization theory (IST), neuronal pathways adapt to respond to this large incentive [2]; thus drug abuse amplifies individuals' desire abnormally and ruins an individual's conception of pleasure [9].

When looking to model gambling addiction from a similar perspective as [2] and [3], we need to first examine whether the concept of pathological gambling satisfies the criteria of addiction in a psychological sense, and thus whether models designed for drug addiction can reasonably be adapted for gambling addiction. In [8], the authors give four criteria which define addiction:

1. a desire or need to continue the activity which places it outside voluntary control,
2. a tendency to increase the frequency or amount of the activity over time,
3. psychological dependence on the pleasurable effects of the activity,
4. a detrimental effect on the individual and society.

Indeed, pathological gambling shares all of these features with drug addiction, and so most current studies treat gambling as a form of addiction [10] [11]. While gambling does not directly induce dopamine release via chemical pathways in the same way as additive substances, the immediate monetary reward from gambling as well as the anticipation of winning money or "beating the system" create a large reward in the brain and accelerate addiction ([12], [8]).

## 2.2 Reward Prediction Error

Reward prediction error (RPE) is a concept which has been heavily used when discussing addiction [13], as well as in mathematical models of addiction [2]. RPE can be defined as discrepancy between the predicted and actual dopamine reward of a stimulus [13], and it embodies reinforcement learning in the neuronal system. It is mediated by dopaminergic neurons which respond differently to rewards which differ from expectations: when the RPE is positive (i.e. the reward is greater than expected), neural activity increases, and otherwise it decreases [2]. Drug addiction distorts the body's perception of dopamine reward, leading to a consistently negative RPE.

Unlike drug addiction where quantitative level of dopamine stimulation can be studied through experiments, to qualitatively determine RPE for pathological gambling, we need to determine which stimulus regulates the release of pleasure in the process. In [8], the author proposed that it was both the monetary reward and the belief of beating the system. [12] also verified that immediacy of reward strongly influences the rate of addiction through experiments. Because the psychological definition of "beating the system" is difficult to quantify, in our modeling we choose the amount of monetary reward as the sole stimulus for dopamine release in gambling.

### 2.3 Opponent Process Theory

Another useful principle for modeling the physiological process of addiction is 'opponent process theory (OPT), which proposes that each emotion, pleasant, or unpleasant, will be followed by a countering response in order to restore homeostasis ([14], [2]). Under this assumption, pleasurable activities like gambling or the consumption of addictive substances release dopamine, which induces a pleasant "a-process" characterized by euphoria, rush, and pleasure. Then a "b-process" defined by withdrawal and craving arises afterwards in order to compensate and restore homeostasis [2]. Continuation of these activities leads to neuroadaptation, where the "b-process" becomes more prevalent and takes place earlier. Tolerance under neuroadapatation made the unpleasant "b-process" dominant [15]. Mathematically, the process of neuroadaptation can be represented by changes in model parameters which describe the dopamine, a-process, and b-process. Eventually, the user's behavior accelerates the process of addiction and he or she falls into an unpleasant overall experience, with a consistently negative RPE, which is one mathematical definition of addiction.

### 2.4 Hebbian Learning

Another approach uses the Hebbian learning rules to model the development and severity of addiction, specifically gambling. Hebbian Learning was first introduced by Donald Hebb, who proposed a neurophysiological account of learning and memory based on a simple principle: "When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased" [3]. In simple terms, "what fires together, wires together". As the axon of cell A repeatedly triggers cell B, the two axons eventually become intertwined and form a link. The link can be used to represent the association formed between knowledge during the learning and behavior process. We believe the same learning is present when an individual is exposed to a strong stimulus. After repeated exposure to the stimulus, the link between the two stimuli and the reward strengthens, and addiction is formed when the link between the two cells has grown an overwhelming degree.

## 3 Opponent Process Theory Model

### 3.1 Previous Opponent Process Theory Models in the Literature

In this section we will discuss the opponent process theory model for drug addiction in Chou et. al. [2], which served as an inspiration for our opponent process theory model of gambling addiction. [2] first proposes to model $D(t)$, the magnitude of activity of dopaminergic neurons following a drug intake as

$$D(t) = \Delta e^{-\delta t}$$

where $\Delta$ represents the magnitude of the dopamine response, assumed to be proportional to the drug dosage, and $\delta$ represents the rate of dopamine clearance. Experimental measurements show that after an initial short transient period in which it rises to a peak, the dopaminergic neural activity decays over a time scale of one to five hours, suggesting that $\delta$ should lie between 0.2 $\mathrm{h}^{-1}$ and 1.0 $\mathrm{h}^{-1}$. However, Chou abstracts this clearance time away by rescaling time $t \mapsto \delta t$, so that 1 unit of time represents the time for the dopamine level to reach $1/e$ of its peak concentration, which would be around 1 hour to 5 hours, depending on the user and the drug. So $D(t)$ can be said to satisfy the differential equation

$$D'(t) = -1.0D(t), \qquad D(0) = \Delta.$$

Then the levels of neural activity corresponding to the $a$-process and $b$-process are determined by the ODE system

$$w_a'(t) = -\alpha w_a(t) + \Gamma^a D(t),$$
$$w_b'(t) = -\beta w_b(t) - \Gamma^b w_a(t),$$

where $\theta := \{\alpha, \beta, \Gamma^a, \Gamma^b\}$ are parameters which represent the neurophysiology of the user. Note that for a positive dopamine response $D(t)$, $w_a$ will take on positive values and $w_b$ will taken on negative values; the interpretation being positive and negative emotions, respectively. Hence in this formulation the total net dynamic response is $w = w_a + w_b$.

Chou assumes that drug intakes occur at times $T_1, T_2, \ldots, T_k, \ldots$, where $T_k = (k-1)T$, and with dosage amounts $\Delta_1, \ldots, \Delta_k$ [2]. In effect, this means that the evolution $D(t)$ is forced by impulses at the intake times, i.e.

$$D'(t) = -D(t) + \sum_{k=1}^{\infty} \delta(t - T_k)\Delta_k$$

where here $\delta$ represents the Dirac delta function, a unit impulse. In the absence of drug intakes, $D$, $w_a$, and $w_b$ all tend to an equilibrium state of 0.

To model the neuroadaptive changes that occur with addiction, Chou proposes that $\beta$ and $\Gamma^b$ be modeled as piecewise constants functions which take constant values $\beta_i$, $\Gamma_i^b$, respectively, on the intervals $[T_i, T_{i+1})$. The values at the $i$th interval are determined by

$$\beta_{i+1} = (1 - B\Delta_i)\beta_i, \qquad \Gamma_{i+1}^b = (1 + G\Delta_i)\Gamma_i^b$$

where $B, G > 0$ represent neuroadaptive parameters depending on the drug and on the user.

Chou also proposes that the drug dosage can depend on the reward-prediction error, which for the $k$th time period is defined as the total reward for the last drug intake minus the total reward for the penultimate drug intake,

$$\mathrm{RPE}_k = \int_{T_k}^{T_{k+1}} w(t)\,dt - \int_{T_{k-1}}^{T_k} w(t)\,dt.$$

Here the net reward from the penultimate drug intake is used as a stand-in for the user's expectation of net reward for the next intake.

In the case of drug dosage changes depending on the RPE, Chou suggests the updating rule

$$\Delta_{i+1} = \Delta_i + \sigma \begin{cases} 1 & \mathrm{RPE}_i \le -R_c \\ \mathrm{RPE}_i / R_c & -R_c \le \mathrm{RPE}_i < 0 \\ 0 & \mathrm{RPE}_i \ge 0. \end{cases}$$

Chou also proposes a separate model for updating the intake timings, but that is not necessary to consider here.

An example result of Chou's deterministic model, run with the starting conditions $D(0) = \Delta_1$, $w_a(0) = w_b(0) = 0$, can be seen in Figure 1, below.

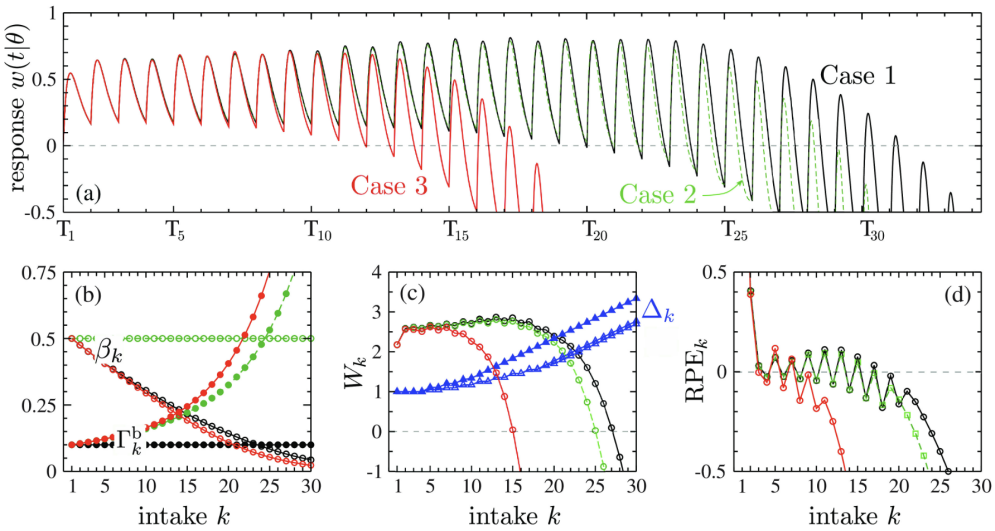



Figure 1: Taken from [2], Figure 4. The time-dependent response resulting from multiple drug intakes with varying $\Delta_k$ for three scenarios. In Case 1 (solid black curve), we set $B = 0.05, G = 0$ while in Case 2 (dashed green curve), $G = 0, B = 0.05$. Finally, in Case 3 (solid red curve), $B = G = 0.05$. The interval between two consecutive intakes in these examples is $T = T_{k+1} - T_k = 6$. The effects of neuroadaptation when both $\beta_k$ and $\Gamma_k^b$ evolve are synergistic as Case 3 leads to addiction after significantly fewer intakes. (b) Evolution of the parameters $\beta_k$ (open circles, solid curves) and $\Gamma_k^b$ (filled circles, dashed curves) for Cases 1, 2, and 3. (c) Evolution of the integrated response and intake doses $\Delta_k$ [blue triangles, Eq. (12)] associated with intake $k$ for each of the three cases. The evolution of $\Delta_k$ is similar for Cases 1 and 2, while $\Delta_k$ for Case 3 rises faster and might describe a highly addictive drug that results in addiction after a smaller number of doses. The integrated responses $W_k$ become negative at about intake $k \approx 27, 26$ and 16 for Cases 1, 2, and 3, respectively. (d) The $\mathrm{RPE}_k$ as a function of the intake $k$ exhibit oscillations with increasing then decreasing amplitude before monotonically decreasing well below zero at intakes $k \approx 20, 18$ and 10 for Cases 1, 2, and 3, respectively. In all cases, the user experiences a "yo-yo" progression to addiction. Since in all cases, $W_k$ becomes negative after the RPE, addiction occurs when $W_k < 0$ at intakes $k_* \approx 27, 26$ and 16, respectively.

### 3.2 An Opponent Process Theory Model for Gambling Addiction

Our model of gambling addiction retains many of the same features as that of Chou et. al. in [2]. In our model, we assume that there is only one time of gambling, namely a slot machine, and that at unit times $t = 0, 1, 2, \ldots$, the user can decide whether or not to gamble by pulling or not pulling the lever. Thus, at each time $k$, the gambler's state, $S_k$, is either $G$ (gambling) or $N$ (not gambling), so that $S_k \in \{G, N\}$ for all $k \in \mathbb{N}$. We assume $S_k$ follows a time-dependent Markov chain with transition probabilities

$$P = \begin{pmatrix} \alpha_G(t) & 1 - \alpha_N(t) \\ 1 - \alpha_G(t) & \alpha_N(t) \end{pmatrix}$$

where $\alpha_G(t), \alpha_N(t) \in (0, 1)$ represent the probability that the individual will continue gambling at the next time step if they are currently gambling, and the probability that the individual will continue to not gamble at the next time step if they are not currently gambling, respectively.

At times $t \in [0, \infty)$ we define the a-process neural activity $w_a(t)$ and the b-process neural activity $w_b(t)$, as well as $D(t)$, the level of dopaminergic activity. Note that these quantities are defined in terms of abstract levels of neural activity. If $S_k = G$, the individual plays a slot machine at time $k$ and receives a stochastic $r_k$, which proportionally increases $D(t)$ (or simplicity, the size of the user's bet is assumed to be constant, so that this reward is drawn from a fixed distribution)

$$\begin{aligned} D'(t) &= -\gamma_d D(t) + \sum_{k:S_k=G} \delta(t - t_k) r_k \\ w_a'(t) &= -\gamma_a w_a(t) + \Gamma^a D(t) \\ w_b'(t) &= -\gamma_b(t) w_b(t) + \Gamma^b(t) w_a(t). \end{aligned}$$

Recall that in [2] the b-process activity level is generically a negative quantity. Here we use the opposite sign convention, so that $w_b$ will generically be positive; hence the total net reward is $w = w_a - w_b$ for our model. We chose this sign convention as it reinforces the intuition that $w_a$ and $w_b$ represent levels of activity in certain neural processes; which should be nonnegative.

In our model we view each slot lever pull as a small part of a gambling experience, which consists of many consecutive pulls. Define $G_j$, $j \geq 0$ and $N_j$, $j \geq 0$ to be the time at the start of the $j$th gambling or non-gambling period, respectively (where $G_0 = 0$):

$$G_j = \min\{t_k : S_k = G; S_{k-1} = NG, t_k > NG_{j-1}\}$$

$$N_j = \min\{t_k : S_k = NG; S_{k-1} = G, t_k > G_j\}.$$

Thus the period from $t = G_j$ to $t = N_j$ represents the $j$th gambling period, which we assume the user remembers as one cohesive gambling experience. Therefore we define the $j$th reward-prediction error by the difference between the total reward for the $j$th gambling experience and the total reward for the $j - 1$th gambling experience:

$$\mathrm{RPE}_j = \int_{G_j}^{N_j} w_a(t) - w_b(t)\, dt - \gamma_r \int_{G_{j-1}}^{N_{j-1}} w_a(t) - w_b(t)\, dt$$

Here, as in [2], the net reward from the penultimate drug intake is used as a stand-in for the user's expectation of net reward for the next intake.

Since the reward distribution in the model, the dominant factors in determining the amount of gambling by the user are the Markov transition probabilities $\alpha_G$ and $\alpha_N$, which affect the number of times (on average) that the user gambles in a row. Thus and $\alpha_G$ and $\alpha_N$ serve a similar purpose to the dosage level in [2]. Therefore, in our model, these probabilities similarly change based on the reward prediction error (RPE). If the RPE is negative, it indicates the user's expectations have not been met, and they will in response increase the probability that they will continue to gamble in the future. Thus, at time $t = N_j$, when the user has just finished a gambling experience, we will update $\alpha_G$ according to the following rule:

$$\alpha_G(N_j) = \alpha_G(N_j - 1) + \sigma \times \begin{cases} 1 & \text{if } \mathrm{RPE}_j \leq -R_c \\ x/R_c & \text{if } -R_c < \mathrm{RPE}_j \leq 0 \\ 0 & \text{if } 0 < \mathrm{RPE}_j \end{cases}$$

where $R_c$ is a threshold value. Furthermore, the neurological parameters $\gamma_b, \Gamma^b$ change over time due to habituation:

$$\gamma_b(t_k+1) = \gamma_b(t_k) \times \begin{cases} \Delta_1 & \text{if } S_k = G \\ \Delta_1^{-1} & \text{if } S_k = N \end{cases}$$

$$\Gamma^b(t_k+1) = \Gamma^b(t_k) \times \begin{cases} \Delta_2 & \text{if } S_k = G \\ \Delta_2^{-1} & \text{if } S_k = N \end{cases}$$

where $\Delta_1 \in (0,1)$ and $\Delta_2 > 1$.

To summarize, the model consists of the continuously varying quantities $D(t), w_a(t), w_b(t)$, the piecewise-constant quantities $S_k, \alpha_G, \alpha_N, \gamma_b, \Gamma^b$, and the fixed quantities $\gamma_d, \gamma_a, \Gamma_a, \Delta_1, \Delta_2, \sigma, R$.

### 3.2.1 Units of Time in our Model

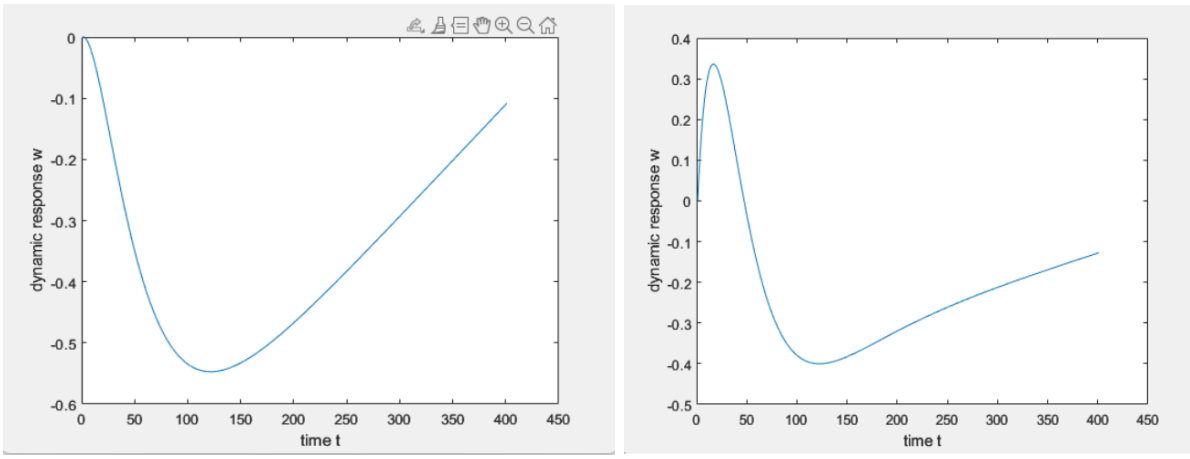
Some caution is in order regarding the units of time and the selection of $\gamma_b$ in our model. We first note that we have chosen our time units so that the interval between opportunities to pull the slot lever represents one unit of time. Thus the unit of time in our model is the number of opportunities the gambler has had to pull the slot lever.

Based on the way we have constructed the model, $1/\gamma_d$ should represent the amount of time for dopaminergic activity to decay by a factor of $1/e$. If we pick $\gamma_d$ using this principle and suppose that dopaminergic activity for gambling decays at the same rate as for addictive substances, then we would expect $\gamma_d$ to be between 0.2 $\text{h}^{-1}$ and 1.0 $\text{h}^{-1}$, which is between $3.33 \times 10^{-3}$ $\text{min}^{-1}$ and $1.66 \times 10^{-2}$ $\text{min}^{-1}$; far too slowly for the level of dopamine to decay appreciably between slot lever pulls which are spaced around one minute apart. For the purposes of our model, we therefore pick $\gamma_d = 1.0$ so that $D(t)$ can decay significantly between slot lever pulls. We partially justify this by noting that a hallmark of gambling is the immediacy of reward, and since gambling does not trigger dopamine-releasing chemical pathways in the same way as addictive substances, it is reasonable to suppose that neural activity following a reward in gambling decays much more quickly than following intake of an addictive substance.

## 3.3 Numerical Simulations

We now introduce some numerical simulations of the model described in the previous section. The plots display qualitative behavior of the net dynamic response $w$ under different fixed values of parameters. The horizontal axis, time $t$, is the number of opportunities the gambler has had to pull the slot lever. The vertical axis represents a unitless measurement of neuronal activity in the $a$ and $b$ processes.

### 3.3.1 Simulation of a Single Bet




(a) Total Dynamic Effect When Losing A Game (b) Total Dynamic Effect When Winning A Game

Figure 2: Simulation of a Single Slot Lever Pull

Unlike addictive substances, a single gambling experience can yield either a positive or negative response depending on whether the user wins or loses money. Consequently, we do not expect to always see a pattern of $w$ being positive at first and becoming negative when "b-process" are triggered. Instead, when losing a bet, the initial "a-process" will start off negative and become gradually smoothed out over time.

Figure 2 shows the dynamic response $w$ under $\gamma_d = 1, \gamma_\alpha = 0.5, \Gamma^\alpha = 1, \Gamma^\beta = 0.8$ and $\gamma_\beta = 0.45$. Figure 2a represents the case where $w$ is always negative, and Figure 12b represents the case where a positive "a-process" triggers the opponent "b-process". Through simulation of one single bet, it is observed that the process of gambling addiction can be more stochastic than drug addiction.

### 3.3.2 Simulation of Single Slot Lever Pull When Varying Parameters

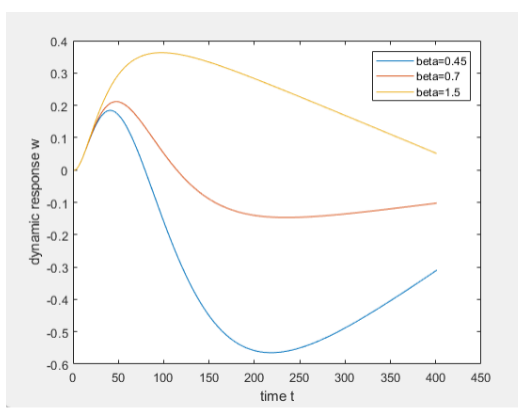



Figure 3: Total Dynamic Effect Under Different Parameter Values

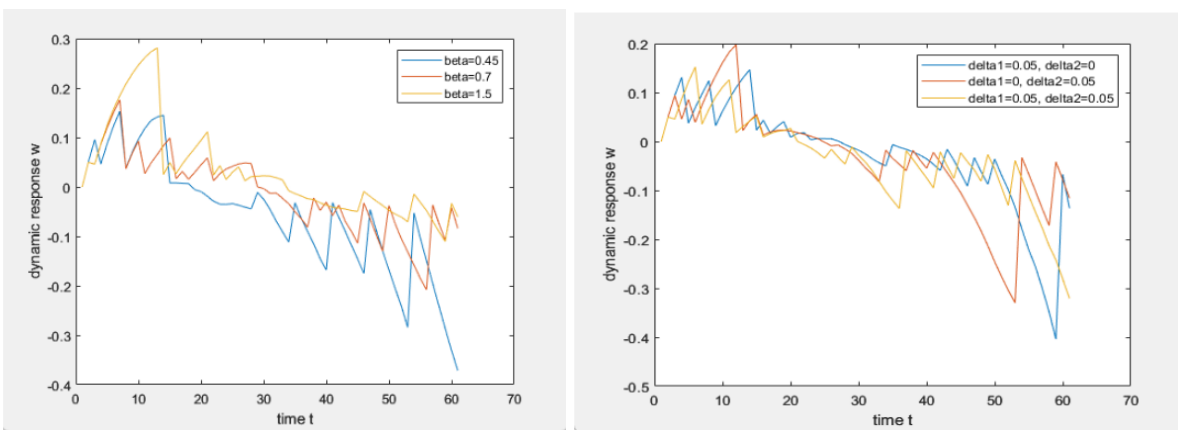
Figure 3 depicts the response curve for players with different parameters $\gamma_\beta = 1.5, 0.7$ or $0.45$ who receive an equal and positive reward. Under varying parameters, the response curve can be (i)

always positive (e.g. $\gamma_\beta = 1.5$), (ii) turning negative with a positive integral (overall experience), e.g. $\gamma_\beta = 0.7$, and (iii) turning negative with a negative integral e.g. $\gamma_\beta = 0.45$.

The three plots display three different types of players. Type (i) represents healthy first-time players who mostly experience pleasure in the process of gambling. For smaller $\gamma_\beta$ or larger $\Gamma^\beta$, players exhibits a type (ii) response as they get more used to the experience. Type (iii) indicates an even stronger "b-process". And the overall experience is negative. Type (ii) and (iii) are more often seen in moderate or addicted gamblers.

Now that we have discussed the simulation for different players in a single slot lever pull, we now start to consider the total response of players in a successive process of gambling and neuroadapatations to model how users move from a type (i) experience to a type (ii) or (iii) experience under changes of parameters.

### 3.3.3 Simulation of Multiple Slot Lever Pulls




(a) Total Dynamic Effect of Multiple Slot Lever Pulls Under Different Parameter Values

(b) Total Dynamic Effect of Multiple Slot Lever Pulls With Neuroadapatation

Figure 4: Multiple Slot Lever Pulls With or Without Neuroadapatation

Figure 4a incorporates the single slot lever pulls modeled in 3 into the Markov chain $P$, allowing the gambler to transition between gambling and not gambling with probability $a_G = 0.5$ and $a_N = 0.7$. In the plot we observe a decreasing trend in the overall level of $w$. Nonetheless, this process occurs at a rather uniform speed, resulting from the inherent structure of the OPT model. But under $\gamma_\beta = 0.45$ with neuroadapatation, the three plots showing $\Delta_1 = 0.05, \Delta_2 = 0$, $\Delta_1 = 0, \Delta_2 = 0.05$ and $\Delta_1 = 0, \Delta_2 = 0$ with $\sigma = 0.05$, $R = 0.5$ all display acceleration in the decrease of $w$ and less duration of the initial pleasant stage, representing the transition towards addiction.

### 3.3.4 Simulation with Dosage Change

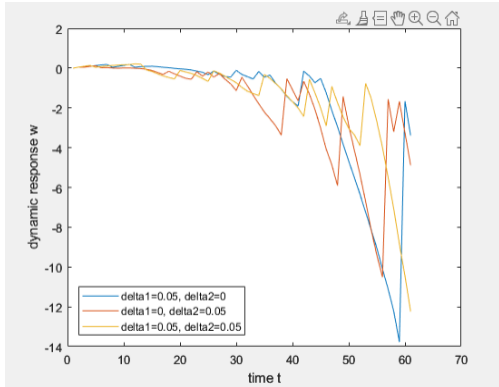
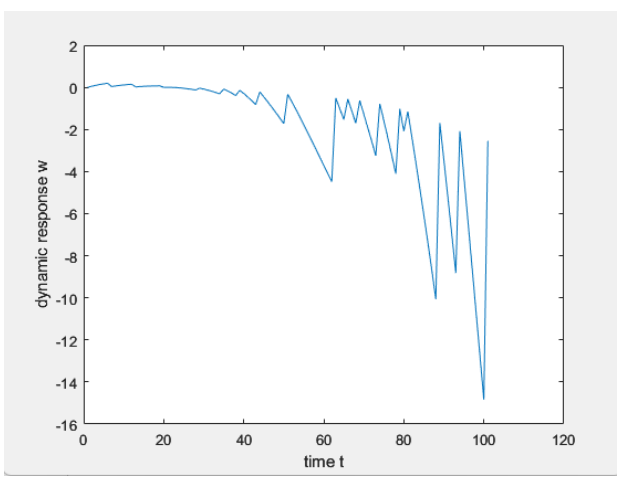



Figure 5: Total Dynamic Effect with Dosage Change

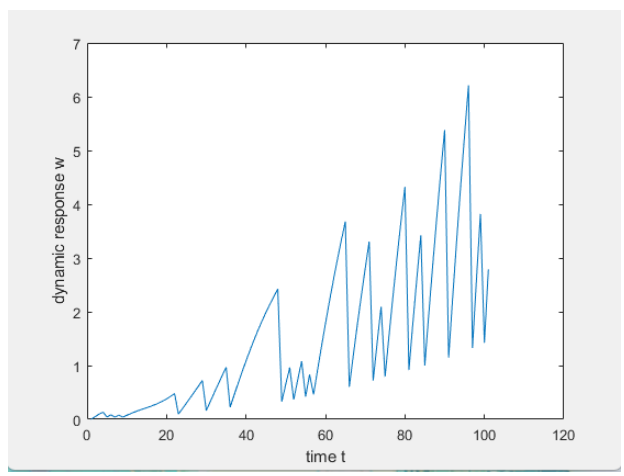
Prior to this section, we assumed that players takes the same dosage (represented by the amount of money they bet) throughout the whole process. But in reality, players may increase their bet when RPE$<0$ and their expected level of reward have not been met. The fluctuation in dosage changes can be more volatile compared to drug addiction, in which case users are limited by biological considerations.

In Figure 5, in addition to reflecting neuroadapatations in parameters, the maximum possible value that players spend in a single bet increases after each bet. And players randomly bet between 0 and the maximum value in each iteration. Here we observed an increasingly exaggerated trend in the decrease of $w$ due to the increase in the level of stimulus. And the addiction process is accelerated at a rapid rate. Therefore, comparing with previous simulations, we can conclude that the evolution of dosage in bets contributes the most to the development of gambling addiction.

### 3.3.5 Averaged Effects




(a) Average Dynamic Effect When Expected Reward $f = -0.2$




(b) Average Dynamic Effect When Expected Reward $f = 0.2$

Figure 6: Averaged Total Dynamic Effect

The gambling process is random and difficult to predict. Slot machines and coin flips are often modeled with a Bernoulli distribution of some form. However, during the simulation process, each run will produce varying results due to the stochastic components of the models, especially as the number of probabilistic components increase. Thus, it becomes critical to average or aggregate the random functions used in the model. Figure 6 demonstrates the average dynamic effect when the reward function is averaged. We can observed that the average of a reward function also plays an important role in the addiction process. And with positive expected returns (It's more likely to win than to loose), $w$ remains positive in the whole process. Therefore, the form of reward function $f$ is also an important factor in OPT model. And in future work we may continue to explore the form of $f$ and its association in the addiction mechanism.

As a further demonstration of system trajectories under a slightly different gambling function, we produced the following plots (Figures 7 and 8), where the slot gives reward $+1$ with 50% probability and $-0.63$ with 50% probability. The parameters used in these plots are $\gamma_d = 1.0, \gamma_a = \Gamma_a = 1.0$, $\Delta_1 = 0.99$, $\Delta_2 = 1.01$, $\sigma = 0.02$, $R_c = 1.0$, $\alpha_N(0) = 0.9$, $\alpha_G(0) = 0.8$.

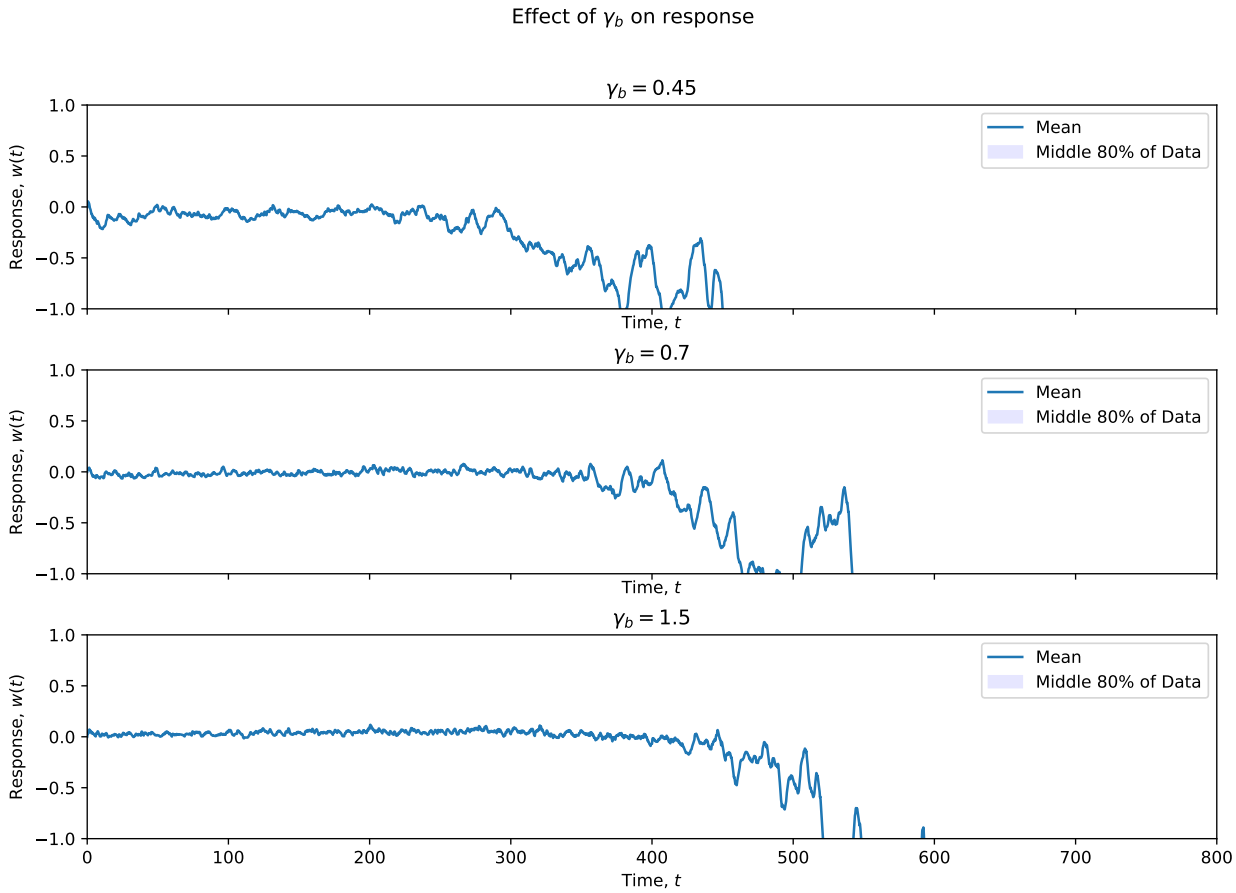



Figure 7: Mean trajectory and middle 80% interval for $\Gamma^b(0) = 1.0$ and $\gamma_b(0) = 0.45, 0.7, 1.5$. We can see that a smaller starting value for $\gamma_b$, indicating a more persistent $b$-process, leads to quicker addiction on average. As $\gamma_b$ increases, the range of outcomes grows smaller due to quicker decay of the $b$-process, and addiction is slowed as well.

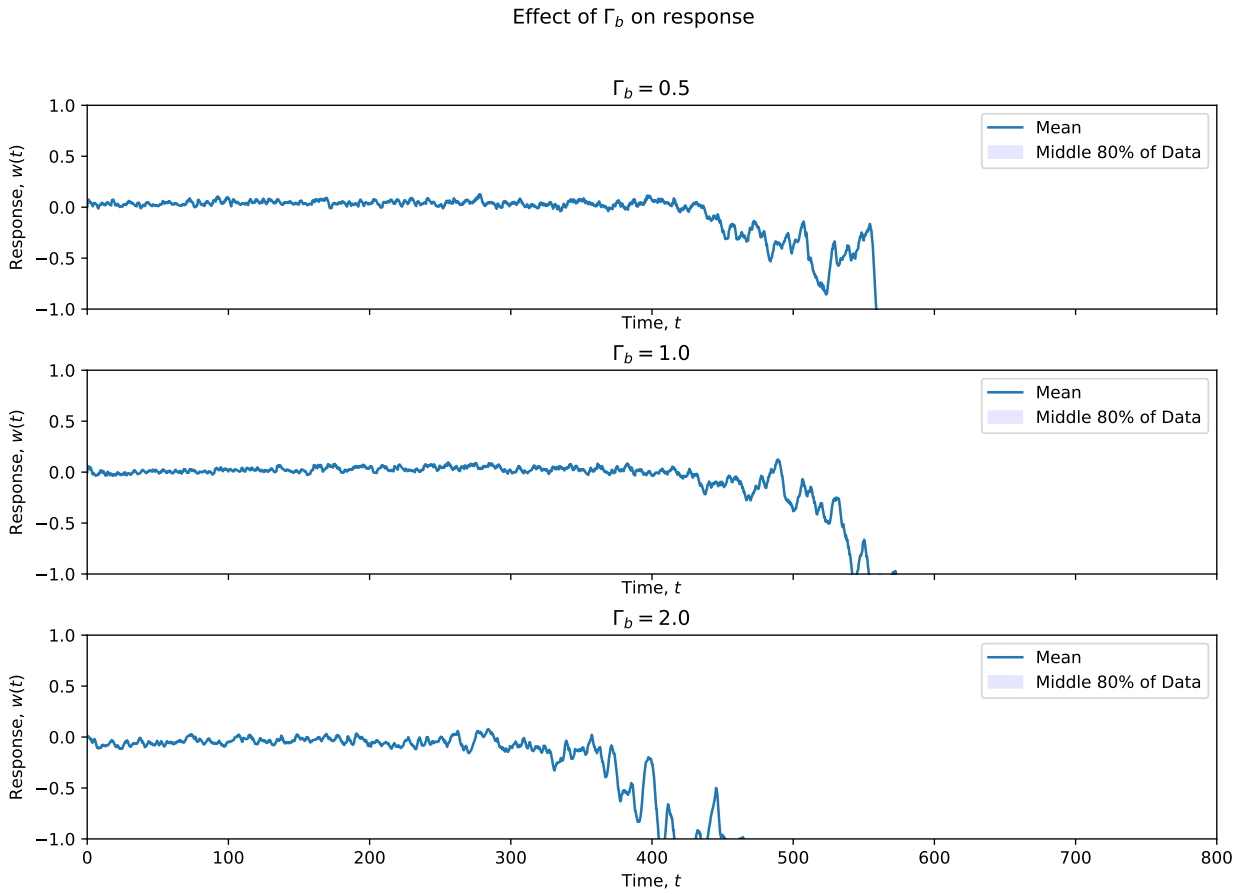



Figure 8: Mean trajectory and middle 80% interval for $\gamma^b(0) = 1.0$ and $\Gamma_b(0) = 0.5, 1.0, 2.0$. We can see that a smaller starting value for $\Gamma_b$, indicating a weaker coupling between the a and b processes, leads to slower addiction on average. As $\Gamma_b$ increases, the speed of addiction is quickened on average.

### 3.4 Conclusion

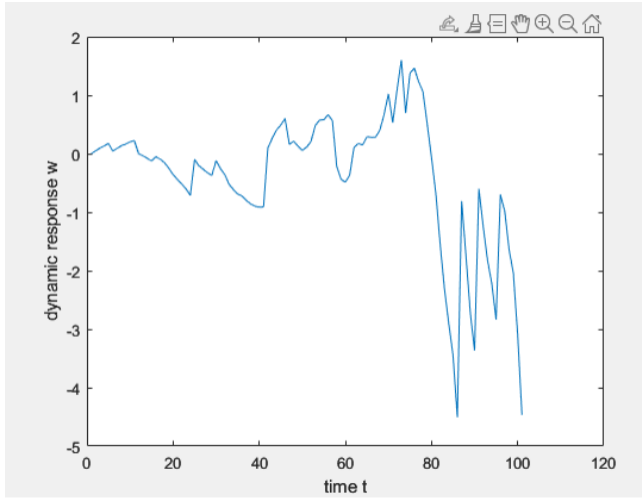



Figure 9: Special Case due to a Stochastic Reward

In this section, we constructed a quantitative model for the evolution of gambling addiction using OPT and RPE. The numerical simulation provided some insight into the addiction process. We can observe that dosage change based on the amount of money players bet is the most influential factor in the mechanism. And the process of neuroadapatation, which influences parameters in the system, can also cause the type (i) dynamic effect curve to gradually evolve into type (ii) or (iii), leading to a negative overall experience.

Different from modeling drug addiction, the process of gambling addiction can be more stochastic because the reward can be negative in some cases. Influenced by the stochasticity in reward function, simulations also vary in each run. Under extreme circumstances showed in Figure 9, it is also possible that the overall dynamic effect remains positive for a while after the addiction mechanism has been triggered. Nonetheless, this also increases player's RPE and leads to an increased tendency to stay gambling afterwards.

As a result, it is also worthwhile to study the distribution of the reward function and its effect on the addiction process, since the reward function can largely influence net dopamine release and plays a deterministic role in the model. Such investigations can be conducted in future work.

## 4 Hebbian Learning Model

### 4.1 Historical Application of Hebbian Learning

Hebbian learning rules have been used repeatedly to study the development of biological systems and language learning. Physiological and biological evidence supports that Hebbian learning rules apply to the strengthening of synaptic efficacy when exposed to impulses. One specific example that follows the Hebbian learning rule is synaptic efficacy within the brain. Research on long term potentiation, conducted on the hippocampus of rats, suggests that after repeated exposure to impulse can lead to longer lasting post-synaptic responses with weaker signals. When exposed to the initial large stimuli, a large reaction is observed, followed by a short delay afterward. Overtime, the efficacy of the response is increased along with the duration. The Hebbian learning rules can be observed from the behaviors of the cells. However, most existing models do not aim to capture the

activities on a neurological level. Instead, they are focused on the system level, such as visual or auditory systems, and behavioral phenomena. McClelland suggests that if the learning rules follow the template model then the relation is considered to be Hebbian [16].

$$\Delta\omega_{rs} = \varepsilon f_r(a_r) f_s(a_s)$$

$\omega_{rs}$ is the weight of the synapse, or strength of link, between the two cells. $\varepsilon$ is the learning rate. At each update interval, the weight is updated with the multiplication of the presynaptic event $f_r(a_r)$ and the postsynaptic event $f_s(a_s)$. A basic model consists of having $f_r(a_r)$ and $f_s(a_s)$ as identity functions, and $a_r$ and $a_s$ takes the values 1 or 0 depending on the activation state. This model presents the Hebbian learning rule in a simple and straightforward method. When both cells are activated at the same time, the weight will increase by $\varepsilon$. However, while it captures the relation, this aforementioned model fails to capture the full complexity of biological systems, learning behaviors, or the long term potentiation. Further research regarding long term potentiation suggests that the postsynaptic event should be dependent on the current existing weight $\omega_{rs}$, resulting in

$$f_r(a_r) = a_r - \omega_{rw}$$

[16]. The key takeaway from this discovery is utilizing the current weight as feedback to determine the level of postsynaptic activation level.

### 4.2 Gambling Addiction for Hebbian Learning

To start, we must define what does gambling and addiction entails. Gambling involves the risk of loss in return for a greater reward. Addiction is the uncontrollable behavior to continue involvement despite knowing the adverse consequences. Thus, gambling addiction is the continued behavior of risking valuable assets despite continuous loss. Most of the time, gambling addictions form due to the high spike in reward, either both physical, psychological , or emotional, gained from winning. Individuals will often seek out the same high of winning despite knowing the potential, and often unavoidable risk of losing. Previously discussed OPT models take a biological approach and focus on the varying level of dopamine during gambling. In those models, the individual experiences a spike of dopamine, followed by a decrease in level. Addiction is proposed to form when the expected value of dopamine no longer offsets the negative reward process. On the other hand, Hebbian learning looks to capture the development of gambling addiction over time. By treating gambling addiction as a form of synaptic link development, the addiction development can be modeled by the previous proposed model. If the gambler continues to encounter high rewards from gambling, they will be more incentivized to continue despite the potential downside of losing their money in the future. The inverse is also true. If the individual experiences a series of losses, they should develop a negative connotation with the gambling process. By pairing the act of gambling with the reward process, individuals will develop their own association between the two events. A strong, positive synaptic weight is associated with loving (or addicted) to the gambling process.

### 4.3 Assumptions of Hebbian Learning Models

Gambling comes in various forms such as sports betting, lottery tickets, bingo, and black jack. It will be too complicated to model each gambling form in the perspective of Hebbian learning. For this paper, we assume the individual is gambling inside a casino with only slot machines and coin flips.This allows us to simplify the chances of winning and losing to a simple probabilistic

distribution. They have never gambled before and have no previous connotation of gambling. Thus, they are learning how to gamble from scratch and have no existing synaptic link between gambling and the reward process.

Due to the environment and their naiveness, the individual starts off with a 50 percent chance to gamble when presented the opportunity. We also assume that the individual is gambling inside a vacuum, where no outside forces affect their decision to continue. The individual is unfortunately trapped inside the casino and will never be able to leave. As a result, all they can do is gamble. At each time interval, the individual is presented with a chance to gamble. We further assume there is no change in synaptic weight due to time passing. Any changes to the synaptic weight occurs when the user is presented with the opportunity to gamble. This assumption simplifies the model further and puts the Hebbian update interval on a discrete timeline and allows us to use a step by step update contrary to a continuous time frame.

Most importantly, we assume the individual has infinite money and can continue to gamble indefinitely. The only loss they experience is from the gambling machine and exists only to affect the synaptic weight. This is an unrealistic assumption since in real life, there is a finite number of financial resources for the user to gamble. However, this decision is made as we treat gambling addiction as the synaptic weight between gambling and the reward process and intend to focus more on the development portion.

### 4.4 Model Description

All the following models follows the Hebbian learning template model of

$$\Delta\omega_{rs} = \varepsilon f_r(a_r) f_s(a_s)$$

. The individual is exposed to multiple opportunities to pull the lever over time. The change in the synaptic is plotted overtime. It is essential to identify the (or lack of) units for the following graphs. On the x-axis, the time is not associated with traditional temporal units such as seconds or days. Instead, it is marked between the individual's opportunity to gamble. On the y-axis marks the weight of the synaptic weight, which is unitless. Instead, for comparison, the maximum observed synaptic weight through all models is approximately 250 over 1000 opportunities to gamble.

In addition, the models all share a similar reward process. The reward process of gambling can be modeled by two stages: whether you win or not and the amount reward for winning. Since the game is restricted to slot machines and coin flips, the probability of winning is modeled after a simple Bernoulli distribution. The win amount is modeled a lognormal distribution, where the value vaguely represents the monetary gain from winning. When the individual wins, they have a high probability of winning a small amount and rarely a jackpot. The higher the winning amount, the stronger the synaptic link formed.

The learning weight, different from synaptic weight, aims to capture how much information a user can absorb at a given point in time. It is inversely proportional with the current synaptic weight between the two cells as seen in figure 10. The learning aims to limit the change in synaptic weight as the user forms their own opinion. The higher the synaptic weight, the less synaptic weight changes. This behavior aims to capture the phenomena that individuals will absorb and be less impacted on the 500th gambling interaction compared to the 1st.

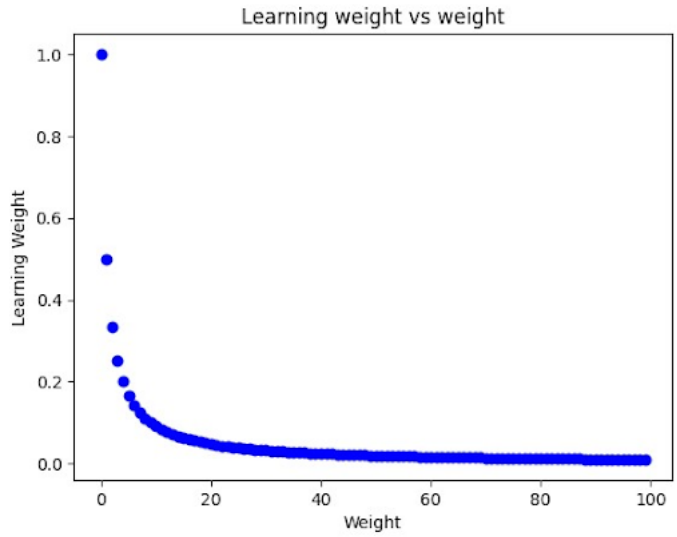



Figure 10: Synaptic weight (x-axis) vs Learning Weight (y-axis)

### 4.5 Model Iteration

The first iteration of the model aims to capture how the reward process and learning rate affect the synaptic weight over time. It assumes the user continuously gambles regardless of the situation. In this model, the user only receives a loss of 0 when losing. Thus, the following Hebbian learning model is proposed. Various probability of winning are tested and plotted in figure 11. From the figure, it becomes evident that the learning weight does eventually reduce the rate of growth. In all cases, the individual eventually forms a positive synaptic with the magnitudes directly dependent on the winning rate.

$$f_r a_r = 1$$

$$f_s a_s = \frac{1}{1+\omega} * Bern(a_s) * LogNormal(\mu_x, \sigma^2)$$

$$\varepsilon = 1$$

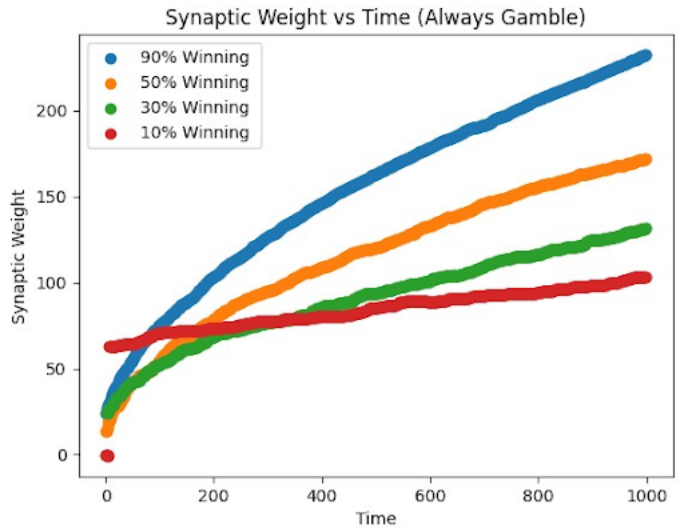


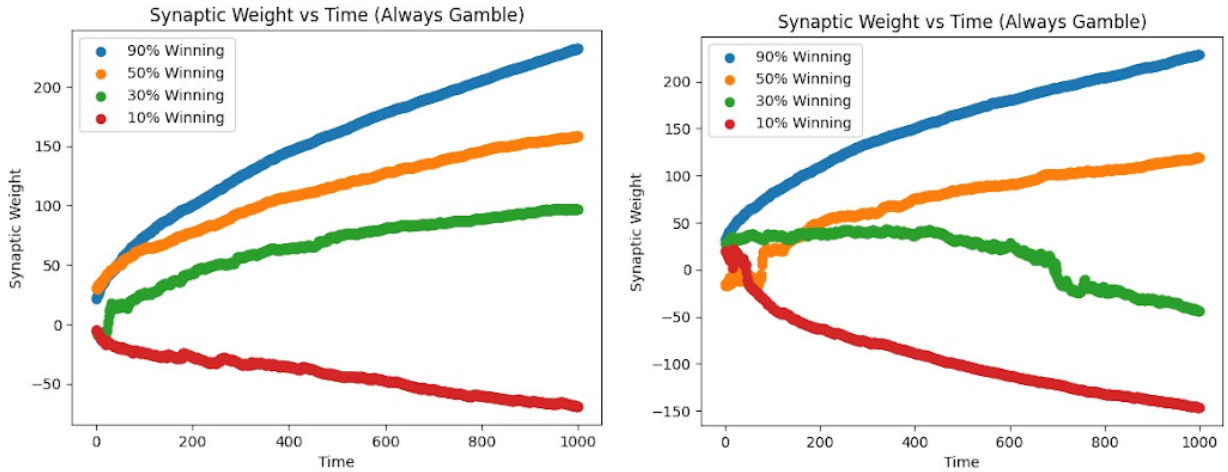

Figure 11: The individual always choose to gamble at varying probability of winning.

The first iteration of the model fails to capture the loss associated with gambling as it does not treat loss in a negative manner. Unlike learning a new language or development of a biological system, the reward process of gambling does not strictly increase. Instead, it fluctuates based on the wins and the losses. In the development of gambling addiction, a loss should result in a negative association, thus negative $\omega$ instead. Thus, in the second iteration, the post synaptic function is adjusted to include a negative value when the Bernoulli distribution is 0.

$$f_s(a_s) = \frac{1}{1+\omega} * f_{reward}(a_s)$$

$$f_{reward}(a_s) = \begin{cases} LogNormal(\mu_x, \sigma^2) \text{ if } Bern(a_s) = 1 \\ -\alpha * LogNormal(\mu_x, \sigma^2) \text{ otherwise} \end{cases}$$

A ratio $\alpha$ is associated with the negative loss. Two $\alpha = .2$ and $\alpha = 1$ are simulated. Two observations are made and presented in figure 12. First, the development of gambling addiction is no longer smooth, especially for trials with low probability of winning. Second, low probability of winning eventually leads to a negative correlation between gambling and winning. An individual can start off with a positive synaptic weight initially but ends up developing a strong negative synaptic weight. By comparing the two alpha, it becomes evident that the scale of the loss also impact whether the individual will develop a positive synpatic weight toward gambling addiction.




(a) $\alpha = .2$

(b) $\alpha = .2$

Figure 12: The individual always choose to gamble at varying probability of winning. But with each loss, they suffer a loss scaled by $\alpha$

In the third iteration, two additional mechanisms are introduced for the fourth iteration: probability of gambling and forgetting. Previous models indicate that the user will continue to gamble regardless of the situation. In the fourth iteration, the user will stop and reconsider their decision to gamble with a certain probability. The decision to gamble is modeled by a Bernoulli distribution. However, delaying the gambling will eventually result in the same trend.

The second mechanism, forgetting, aims to model the decreasing efficiency of synaptic link over time. You lose what you don't use. However, the more well versed you are in it, the less you will forget. Thus, the magnitude of the forgetting is directly proportional to the learning weight.

$$f_r(a_r) = \begin{cases} 1 \text{ if } Bern(a_r) = \text{individual chooses to gamble} \\ -\beta * \frac{1}{1+\omega} \text{ otherwise} \end{cases}$$

For this paper, the $\beta = 1$ is assumed for simplicity. At each step, if the individual chooses not to gamble, they will forget the reward mechanism instead. We assume that the forgetting rate is never positive since the decision to not gamble should not make the user more likely to gamble. As a result, once the user establishes a negative synaptic weight, they are unlikely to continue redeveloping a positive weight unless they experience a significant reward or a series of wins.

As seen in figure 13 When a series of losses occurs, which happens frequently with low win probability, the individual ends up with a negative weight and no longer gambles. With the forgetting rate, the user eventually gets a negative synaptic weight associated with gambling. At 50% probability of winning, the growth of synaptic weight remains relatively constant. The previous models, even with negative loss, does not fully offset the high amount of reward during the win. However, in this iteration, when the user can choose to not gamble, an additional loss is sustained, further limiting the growth of the synpatic efficiacy.

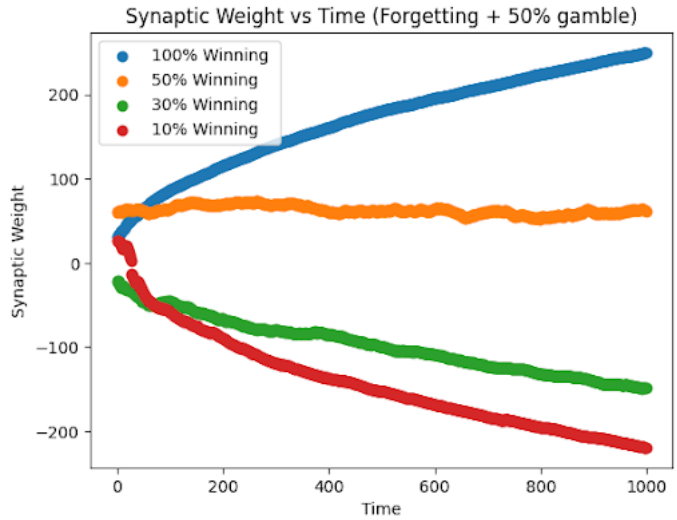



Figure 13: The individual decides when to gamble with 50% probability and will forget if chooses not to

For the final iteration, the probability of gambling is dependent on the synaptic weight is feedback to the decision to gamble or not. The model for the gambling decision is as proposed below. At the first gambling opportunity, the user maintains the 0.5 probability to start gambling. As the synaptic weight increases or decreases, it affects the probability based on the percentage compared to the maximum observed synaptic weight. $max\omega$ is observed to be approximately 250.

$$f_r(a_r) = \begin{cases} 1 \text{ if } Bern(a_r) = \text{individual chooses to gamble} \\ -\beta * \frac{1}{1+\omega} \text{ otherwise} \end{cases}$$

$$a_r = .5 + \frac{\omega}{max\omega}$$

The result of the fourth iteration is significantly more erratic compared to the previous proposed models, especially at low winning rates (Figure 14). While the winning rate is still strongly correlated with the ordering of the synaptic weight, the final sign of the synaptic weight fluctuates between positive and negative. For the 30% winning rate, it is possible for the user to develop a positive correlation before returning to negative. For the 10%, the user can choose to gamble multiple times and experience a spike in synaptic weight when they experience a reward. The unpredictable behavior is dependent on the stochastic model for both the gambling decision and the reward process.

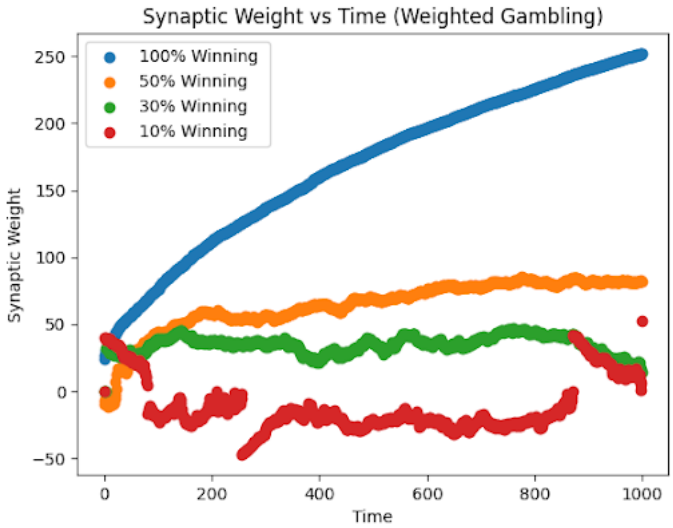



Figure 14: The Individual will choose to gamble, dependent on the existing weight. They can suffer loss of synaptic weight and will forget if they choose not to gamble. However, they will be heavily rewarded when they gamble and win

# 5 Discussion and Future Work

## 5.1 Trade offs between OPT and Hebbian

In this report, we discussed two possible approaches in modeling the gambling addiction process, and provided relevant analysis in quantitative simulations alongside the models.

The proposed models based on Hebbian learning rules provide a simpler mathematical update model compared to the OPT mentioned in the previous section of the paper. On the one hand, Hebbian learning models capture the development of the gambling addiction on a larger scale. On the other hand, OPT dives deep into the biological mechanism and models the process after the dopamine level associated with gambling. The varying dopamine level over time is a strong indicator of addiction. Hebbian models look to capture the process of addiction without diving into the biological details.

Due to the complexity of the OPT model, we had to assume multiple factors compared to the Hebbian model. In addition, the complex model also results in more challenging code and simulations, as it is a continuous-time model. The proposed Hebbian learning models utilize a discrete time scale and updates at set intervals. As a, the Hebbian model struggles to capture more complex behaviors such as reward-prediction error, an essential components for the OPT model.

## 5.2 Future Work

Throughout our iterations, we received numerous feedback on our assumptions. Furthermore, we realized several lacking features throughout our iterations. We document them below.

### 5.2.1 Cost Associated with Gambling

During the final presentation, the question regarding the cost of gambling is brought up. How is the user allowed to gambling indefinitely in such a short period of time? In addition, is it really

an addiction, if there is no cost associated with each gambling choice? We assume that there is no monetary cost of gambling in our models as it will complicate both the gambling decisions, reward processes, loss penalty, and more. Does the user go into debt when they run out of money? Thus, for our paper, we assumed the cost of gambling is associated in the biological sense, either through the dopamine level or the strength of the synaptic link between two cells.

Nevertheless, we agree that integrating monetary values into the gambling addiction models is an essential next step for the study of gambling addiction. There is both an emotional, biological, and monetary component to the addiction after all. By assigning a specific cost to gambling, we can also further explore the relationship between the two.

## References


[1] E. Sohn. "How gambling affects the brain and who is most vulnerable to addiction." (2023), [Online]. Available: `https://www.apa.org/monitor/2023/07/how-gambling-affects-the-brain` (visited on 12/16/2023).

[2] T. Chou and M. R. D'Orsogna, "A mathematical model of reward-mediated learning in drug addiction," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 32, no. 2, 2022.

[3] G. V. Keysers C. "Hebbian learning and predictive mirror neurons for actions, sensations and emotions." (2014), [Online]. Available: `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4006178` (visited on 12/14/2023).

[4] "AGA commercial gaming revenue tracker." (2023), [Online]. Available: `https://www.americangaming.org/resources/aga-commercial-gaming-revenue-tracker/` (visited on 12/16/2023).

[5] "National trends from NGAGE 2.0 (2021)." (2021), [Online]. Available: `https://www.ncpgsurvey.org/2021-national-trends/` (visited on 12/16/2023).

[6] B. Gutkin and S. H. Ahmed, *Computational neuroscience of drug addiction*. Springer Science & Business Media, 2011, vol. 10.

[7] R. Brown, "Models of gambling and gambling addictions as perceptual filters," *Journal of gambling behavior*, vol. 3, no. 4, pp. 224–236, 1988.

[8] M. B. Walker, "Some problems with the concept of "gambling addiction": Should theories of addiction be generalized to include excessive gambling?" *Journal of Gambling behavior*, vol. 5, pp. 179–200, 1989.

[9] G. Di Chiara and A. Imperato, "Drugs abused by humans preferentially increase synaptic dopamine concentrations in the mesolimbic system of freely moving rats.," *Proceedings of the National Academy of Sciences*, vol. 85, no. 14, pp. 5274–5278, 1988.

[10] J. Elster, "Gambling and addiction," *High Culture: Reflections on Addiction and Modernity*, vol. 322, 2003.

[11] P. Mobilia, "Gambling as a rational addiction," *Journal of Gambling Studies*, vol. 9, no. 2, pp. 121–151, 1993.

[12] M. Chóliz, "Experimental analysis of the game in pathological gamblers: Effect of the immediacy of the reward in slot machines," *Journal of Gambling Studies*, vol. 26, pp. 249–256, 2010.

[13] W. Schultz, "Behavioral theories and the neurophysiology of reward," *Annu. Rev. Psychol.*, vol. 57, pp. 87–115, 2006.

[14] R. L. Solomon and J. D. Corbit, "An opponent-process theory of motivation: I. temporal dynamics of affect.," *Psychological review*, vol. 81, no. 2, p. 119, 1974.

[15] O. George and G. F. Koob, "Individual differences in the neuropsychopathology of addiction," *Dialogues in clinical neuroscience*, vol. 19, no. 3, pp. 217–229, 2017.

[16] J. L. McClelland. "How far can you go with hebbian learning, and when does it lead you astray?" (2006), [Online]. Available: `https://stanford.edu/~jlmcc/papers/McClellandIPHowFar.pdf` (visited on 12/14/2023).